First Batch — 73 Non-LLM Agent Cells

Day-One Roster and Build Plan · Wave-1 Re-Baseline

Owner-finalized 2026-10-27. Supersedes the `2–10 cells` day-one sizing in `AGENT-PRODUCTION-BUILD-PLAN.md` §13.

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§0 · Purpose and scope

Item	Value
Cells defined here	**73** (instrument-level)
Venues	**3** — Upstox (NSE/BSE), IC Markets (FX + commodity CFDs), Hyperliquid (on-chain perps)
LLM involvement in these cells	**ZERO** — every cell is deterministic or trained-ML, never an LLM
Symbol validation status	**Verified live** against NSE bhavcopy 26-Oct-2026 (3,526 symbols) and three official NSE index files
What this document is	The day-one cell roster + ordered build plan + gates
What it is not	Not the style roster (30 styles live in `AGENT-TAXONOMY-FAMILIES.md`), not the engine spec

This document replaces the assumption “wave 1 = 2–10 cells”. The owner re-baselined wave 1 to the full 73-cell first batch (decision D1 below). That decision is recorded, not implied.

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§1 · Owner decisions locked this session

#	Decision	Locked value	Consequence
D1	Wave re-baseline	**73 cells IS the new W1**	Wave table in `AGENT-TAXONOMY-FAMILIES.md` §5 and gate C5 in the production report must be rewritten
D2	Commodities trunk	**Hosted on the FX trunk — flagged exception**	The cross-family ban (§3 of taxonomy) gets one named carve-out for Brent, WTI, XAU, XAG
D3	Index cells	**Trade via index F&O** — NIFTY through `NFO`, SENSEX through `BFO`	Pulls futures + options families into day one; needs expiry calendars and lot sizes

D1 changes a documented rule. It does **not** change the underlying engineering constraint: a 73-cell cold start has no precedent in these docs. §9 answers that with tranches inside W1 rather than by reviving the old 2–10 limit.

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§2 · Cell census — 73

#	Trunk	Venue	Cells	Composition
1	**Equities** (cash)	Upstox `NSE_EQ`	**57**	50 NIFTY 50 members + 7 off-index large caps
2	**Futures + Options** (index)	Upstox `NSE_FO` / `BSE_FO`	**2**	NIFTY (NFO), SENSEX (BFO) — underlying-level cells
3	**FX**	IC Markets ECN	**7**	EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, USD/CHF, USD/SGD
4	**FX trunk — commodity exception (D2)**	IC Markets ECN	**4**	Brent, WTI, XAU/USD, XAG/USD
5	**Crypto** (perps)	Hyperliquid L1	**3**	BTC, ETH, HYPE
6	**Predictions**	*none*	**0**	No venue connected — see §11 open decision O1
	Total		**73**	

Count check: `57 + 2 + 7 + 4 + 3 + 0 = 73` ✓

Trunks touched on day one: 5 of 6. Predictions is the only absent trunk, while `AGENT-TAXONOMY-FAMILIES.md` §3 states predictions is an “own trunk from day one”. That contradiction is carried openly to §11 as open decision **O1**.

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§3 · Cell definition — instrument cell vs agent instance

The plan defines a cell as **one symbol × one style** (`AGENT-PRODUCTION-BUILD-PLAN.md` §13). This document counts **instrument cells** (73). The two numbers are different and must not be conflated:

Quantity	Formula	Day-one value
Instrument cells	fixed roster	**73**
Agent instances	instruments × styles per instrument	**73 at 1 style each**
Agent instances if 2 styles each	73 × 2	146
First-decade target (report §13)	liquid-first roster	3,000–6,000

Day-one rule adopted here: 1 style per instrument. Styles multiply later through the normal wave gates. Any document claiming “73 agents” means 73 instances.

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§4 · Symbol master — 57 NSE equities (corrected)

Verified against: `ind_nifty50list.csv` (50 rows), `ind_nifty100list.csv` (100 rows), `ind_nifty500list.csv`, and `sec_bhavdata_full_26102026.csv` (3,526 symbols).

#	Symbol	NIFTY 50	Index tier	Note
1	ADANIENT	✓	NIFTY 50	
2	ADANIPORTS	✓	NIFTY 50	
3	APOLLOHOSP	✓	NIFTY 50	
4	ASIANPAINT	✓	NIFTY 50	
5	AXISBANK	✓	NIFTY 50	
6	**BAJAJ-AUTO**	✓	NIFTY 50	**corrected** — was written `BAJAJAUTO`
7	BAJAJFINSV	✓	NIFTY 50	
8	BAJFINANCE	✓	NIFTY 50	
9	BEL	✓	NIFTY 50	
10	BPCL	—	NIFTY Next 50	off-index (kept by owner)
11	BHARTIARTL	✓	NIFTY 50	
12	BRITANNIA	—	NIFTY Next 50	off-index (kept by owner)
13	**BSE**	✓	NIFTY 50	**corrected** — was written `BSELTD`
14	CIPLA	✓	NIFTY 50	
15	COALINDIA	✓	NIFTY 50	
16	DIVISLAB	—	NIFTY Next 50	off-index (kept by owner)
17	DRREDDY	✓	NIFTY 50	
18	EICHERMOT	✓	NIFTY 50	
19	ETERNAL	✓	NIFTY 50	ex-Zomato
20	GRASIM	✓	NIFTY 50	
21	HCLTECH	✓	NIFTY 50	
22	HDFCBANK	✓	NIFTY 50	
23	HDFCLIFE	✓	NIFTY 50	
24	HEROMOTOCO	—	NIFTY 500 only	off-index (kept by owner)
25	HINDALCO	✓	NIFTY 50	
26	HINDUNILVR	✓	NIFTY 50	
27	ICICIBANK	✓	NIFTY 50	
28	INDIGO	✓	NIFTY 50	
29	INDUSINDBK	—	NIFTY 500 only	off-index (kept by owner)
30	INFY	✓	NIFTY 50	
31	ITC	✓	NIFTY 50	
32	JIOFIN	✓	NIFTY 50	
33	JSWSTEEL	✓	NIFTY 50	
34	KOTAKBANK	✓	NIFTY 50	
35	LT	✓	NIFTY 50	
36	M&M	✓	NIFTY 50	`&` needs URL/JSON escaping

#	Symbol	NIFTY 50	Index tier	Note
37	MARUTI	✓	NIFTY 50	
38	MAXHEALTH	✓	NIFTY 50	
39	NESTLEIND	✓	NIFTY 50	
40	NTPC	✓	NIFTY 50	
41	ONGC	✓	NIFTY 50	
42	POWERGRID	✓	NIFTY 50	
43	RELIANCE	✓	NIFTY 50	
44	SBILIFE	✓	NIFTY 50	
45	SBIN	✓	NIFTY 50	
46	SHRIRAMFIN	✓	NIFTY 50	
47	SUNPHARMA	✓	NIFTY 50	
48	TATACONSUM	✓	NIFTY 50	
49	TATASTEEL	✓	NIFTY 50	
50	TCS	✓	NIFTY 50	
51	TECHM	✓	NIFTY 50	
52	TITAN	✓	NIFTY 50	
53	**TMPV**	✓	NIFTY 50	**corrected** — was written `TATAMOTORS` (PV demerger)
54	TRENT	✓	NIFTY 50	
55	ULTRACEMCO	✓	NIFTY 50	
56	UPL	—	NIFTY 500 only	off-index (kept by owner)
57	WIPRO	—	NIFTY Next 50	off-index (kept by owner)

Coverage result: after the three corrections, the roster contains **100% of the live NIFTY 50** (50 of 50). The 7 off-index names are valid, tradable, and deliberately retained — so the roster must be labelled **“NIFTY 50 + 7 large caps”**, never “NIFTY 50”.

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§5 · The three symbol corrections

Each would have produced a rejected order or a wrong-instrument fill.

Written by owner	Live NSE symbol	Proof — 26-Oct-2026 bhavcopy	Cause
`BAJAJAUTO`	**`BAJAJ-AUTO`**	`BAJAJAUTO` = 0 hits · `BAJAJ-AUTO` = 1 hit	missing hyphen
`BSELTD`	**`BSE`**	`BSELTD` = 0 hits · `BSE` = 1 hit	vendor suffix
`TATAMOTORS`	**`TMPV`**	`TATAMOTORS` = 0 hits · `TMPV` = 1 hit	Tata Motors PV demerger

`M&M` and `ETERNAL` were checked separately and are **correct as written** (1 hit each).

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§6 · Venue matrix

	Upstox	IC Markets	Hyperliquid
Segments	`NSE_EQ`, `NSE_INDEX`, `NSE_FO`, `BSE_EQ`, `BSE_INDEX`, `BSE_FO`, `CDS`, `BCD_FO`, `MCX_FO`, `NSE_COM`, `NCD_FO`	MT4 / MT5 / cTrader / TradingView	on-chain L1, perps only
Cells used	59	11	3
Instrument key format	`NSE_EQ|<ISIN>`, `NSE_INDEX|Nifty 50`, `BSE_INDEX|SENSEX`	broker symbol (`XAUUSD`, `XTIUSD`, …)	bare name (`BTC`, `ETH`, `HYPE`)
Counterparty	Regulated exchange (NSE / BSE)	**CFD issuer — own book**	On-chain, no broker
Cost	brokerage + STT + exchange charges + stamp	Raw Spread: **$7/lot round-turn** (MT) or **$3 per $100k/side** (cTrader)	venue fee + funding
Measured spread	n/a	EUR/USD raw **0.32 pip** avg 24h; XAUUSD **$0.082**	n/a
Hours	NSE 09:15–15:30 IST	~24/5, but **gold 01:02–23:57 server time**	24/7
Overnight cost	NSE F&O carry	XAUUSD long **−$53.31/lot/night**; Brent **+15.74 / −26.24**	funding, floor **0.00125%/8h**

Counterparty asymmetry is material. IC Markets states it *“is the issuer of the products it provides”* and holds a securities-dealer licence permitting it to make a market. FX loss is the broker’s book. Upstox risk clears through an exchange; Hyperliquid clears on-chain. The governor (E5) must therefore carry a **venue risk class**, not a flat exposure number.

Symbol ambiguity to resolve before any oil order: IC’s commodity spec sheet lists Brent as `XBRUSH` and WTI as `XTIUSD`; their commodities page lists `XBRUSD`; a third-party live feed lists `BRENT` / `WTI`. IC’s own disclaimer says the authoritative list is what appears *in the account*. Resolve from the live symbol master at onboarding — never hardcode.

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§7 · The non-LLM guarantee

This batch is the “non-LLM agents” batch. The constraint is structural, not stylistic:

Layer	Allowed	Forbidden
Cell decision (hot path)	deterministic rules, trained ML exported to ONNX	**any LLM call**
Risk / governor (E5)	deterministic arithmetic only	**any LLM call**
Execution (E6)	deterministic order logic	**any LLM call**
Research / cockpit	LLM assistants permitted	must never write into the hot path

Rationale, measured rather than asserted: LLM decision cycles run 15–40 s at ~$0.12 per decision with <5% signal value at a 20 s horizon (MethodAlgo 2026, cited in report §19). Every cell above must decide in µs–ms. An LLM in that loop is not a slow agent, it is a broken one.

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§8 · Build plan — ordered, gated

Steps are dependency-ordered. A step does not start until the one it depends on has passed its gate.

#	Step	Deliverable	Done-when
1	Fix symbol master	corrected 57-symbol roster	3 renames applied; validation gate passes
2	Resolve commodity exception (D2)	taxonomy carve-out clause	Brent/WTI/XAU/XAG named as FX-trunk tenants
3	Resolve index F&O typing (D3)	NIFTY→NFO, SENSEX→BFO cell spec	lot size, expiry calendar, contract-selection rule written
4	Write `VENUE-CONNECTIVITY.md`	3-venue matrix (closes remainder-map item #6)	all cells mapped to a segment + instrument key
5	Symbol-master service	loader + reject-unknown gate	0 unmatched symbols against live Upstox master
6	**IC Markets connectivity decision**	cTrader Open API **or** MT5 bridge	one path chosen and proven with a read-only quote
7	Hyperliquid adapter	WS feed + `meta` universe sync	`BTC`, `ETH`, `HYPE` resolve; `BTC/USDC` rejected
8	Upstox adapter	REST + WS, equity + F&O	quote + depth for 1 equity and 1 index option
9	IC adapter	quote + depth for 1 FX pair + 1 commodity	`XBRUSD` / `XBRUSH` / `BRENT` resolved to exactly one
10	Data plane wiring	NATS bus + hot store ingestion	3 venues streaming, clock skew recorded
11	Cell bootstrap	73 cells instantiated, 1 style each	governance skeleton + risk caps applied
12	Reconciliation + governor	nightly broker-statement recon, venue risk classes	mismatch ⇒ trading pauses (per `AGENT-17-TOPICS-DESIGN.md` §C1)

Steps 6–9 are read-only. **No order is placed until step 12 passes.**

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§9 · Tranches inside W1

The owner re-baselined the wave; the physics of a cold start remain. 73 cells sharing one governor, three venues, three counterparty classes and three clocks is not a single deployable unit. W1 therefore ships in three tranches — this is execution sequencing inside W1, not a return to the old 2–10 limit.

Tranche	Contents	Cells	Gate to pass before the next tranche
T1	Venue connectivity (steps 4–10) + 10 equity cells, 1 style	10	5 consecutive sessions, ledger matches broker, zero stale feeds
T2	Remaining 47 equity cells + 2 index F&O cells	49	governor exposure caps hold under all 3 counterparty classes

Tranche	Contents	Cells	Gate to pass before the next tranche
T3	7 FX + 4 commodity + 3 crypto cells	14	swap/funding costs modelled in the backtest and matched live

Cumulative: `10 → 59 → 73`. Each gate is measured against the ledger, not against intent.

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§10 · Risk register

ID	Risk	Severity	Evidence	Mitigation
R1	MT5 automation is Windows-only; the FX adapter may be unbuildable as assumed	**High**	known limitation	Prefer **cTrader Open API** (WebSocket/FIX); decide at step 6
R2	Counterparty asymmetry — IC is a CFD issuer, not an exchange	**High**	IC help-centre wording	Venue risk class in E5; per-class exposure caps
R3	73-cell cold start exceeds any documented precedent	**High**	report §13	T1→T2→T3 tranche gates (§9)
R4	Three broken symbols → rejected or misrouted orders	**High**	bhavcopy 0-hit proof	Validation gate at step 5
R5	Brent symbol ambiguous across three sources	**Med**	`XBRUSH` / `XBRUSD` / `BRENT`	Resolve from live account master (step 9)
R6	IC hours ≠ 24/5 and swaps are large	**Med**	XAUUSD −$53.31/lot/night	Swap schedule inside the style cost model; no overnight carry for short-horizon styles
R7	Hyperliquid funding pinned at floor — carry ≈ 0 today	**Med**	live: BTC 0.00125%/8h, ETH 0.0000517%/8h	Measure regime before sizing the funding cell
R8	FX has no live volume tape	**Med**	BIS survey only	Name the tick-activity proxy in the E1 feature spec, or buy a feed
R9	Market-making archetypes cannot run on these venues	**Med**	external hop 0.5–50 ms (`DATA-LAYER` §61–64)	MM styles stay deferred
R10	Jurisdiction: IC and Hyperliquid are not US-available; Upstox is India-resident	**Med**	platform policy	Document as single-owner assumption
R11	NIFTY 50 reconstitutes semi-annually — roster silently drifts	**Low**	today’s 7-name diff	Weekly membership refresh + explicit allowlist
R12	Upstox market-data entitlement / rate limits unverified	**Low**	not tested	Test at step 8

ID	Risk	Severity	Evidence	Mitigation
R13	`M&M` breaks URL/JSON handling	**Low**	symbol confirmed live	Escape at the adapter boundary; add a test fixture

§11 · Open decisions

ID	Question	Why it blocks
O1	Predictions trunk has **0 cells**, yet taxonomy §3 says “own trunk from day one”. Add Polymarket/Kalshi cells, or formally defer?	Only remaining wave-rule contradiction after D1
O2	cTrader Open API vs MT5+Wine for IC Markets	Decides the whole FX adapter shape (step 6)
O3	Index F&O contract selection — one underlying cell per index, or per-expiry/per-strike cells?	A “NIFTY cell” could be thousands of instruments; measured 40,212 F&O contracts in one session
O4	Keep the 7 off-index names or replace with true NIFTY 50 members?	Roster label and index-drift monitoring depend on it
O5	First style activated per instrument (currently assumed: 1 style each)	Determines whether day one is 73 or 146 instances

§12 · Verification checklist

Gate	Check	Pass criteria
V1	Symbol master join vs live Upstox master	0 unmatched; `M&M` resolves
V2	Index membership refresh (weekly)	roster ⊇ NIFTY 50, else gaps are in the allowlist
V3	Hyperliquid universe sync	`BTC`, `ETH`, `HYPE` present; `BTC/USDC` rejected
V4	IC symbol resolution	Brent resolves to exactly one live symbol
V5	Adapter heartbeats (3 venues)	all return; clock skew logged; 0 auth errors
V6	Wave gate	73 cells documented as W1 per D1, with tranche gates
V7	Cost baseline	1 round-turn per venue vs published: EUR/USD 0.32 pip, $7/lot RT
V8	Reconciliation	nightly statement match; mismatch pauses trading
V9	Non-LLM audit	0 LLM calls reachable from cell/governor/execution code paths

§13 · Provenance — what was measured, not assumed

All figures in this document were pulled live on 2026-10-26/27 unless marked otherwise.

Claim	Source
57 symbols resolve on NSE	`archives.nseindia.com/products/content/sec_bhavdata_full_26102026.csv` (3,526 symbols)
50 live NIFTY 50 members	`archives.nseindia.com/content/indices/ind_nifty50list.csv` (50 rows)
4 names sit in Next 50	`ind_nifty100list.csv` (100 rows) minus NIFTY 50
3 names sit in NIFTY 500 only	`ind_nifty500list.csv`
Upstox segment codes	Upstox Developer API instrument documentation
`NSE_INDEX|Nifty 50`, `BSE_INDEX|SENSEX` are quote instruments	Upstox instrument master
IC Raw Spread cost	IC Markets trading-costs pages
IC spreads (EUR/USD 0.32 pip; XAUUSD $0.082)	live IC spread feed, 26-Oct-2026
IC swap rates	same live feed, 26-Oct-2026
Hyperliquid perp names + funding	`POST api.hyperliquid.xyz/info {"type":"meta"}` / `metaAndAssetCtxs`
NIFTY lot size 65	F&O bhavcopy field `NewBrdLotQty`
40,212 F&O contracts in one session	F&O bhavcopy 23-Oct-2026
LLM decision economics	report §19, MethodAlgo 2026

Not verified: SENSEX lot size and BSE expiry calendar (blocked: `bseindia.com` returns HTTP 403 to automated requests; `api.bseindia.com` returns *Access Denied*). Must be read from the Upstox instrument master at step 3.

Companion to `AGENT-PRODUCTION-BUILD-PLAN.md` (engines, orchestration, population) and `AGENT-TAXONOMY-FAMILIES.md` (6 trunks, 30 styles, waves). Session 24 in `planing conva/PLANNING-CONVERSATION-LOG.md` records the decisions D1–D3.